BIKE SHARING STATION PERFORMANCE AND UTILIZATION ANALYSIS

1. Project Overview and Objective

This project focuses on analysing bike-sharing station data using Power BI. The raw dataset was first cleaned and transformed using Power Query, followed by data modelling and DAX calculations in Power BI.

The objective of this project is to build an interactive Power BI dashboard that helps analyse station activity, bike availability, and utilization patterns across cities and stations.

The dashboard enables users to understand operational efficiency, identify underperforming locations, and monitor bike station performance through visual insights.

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2. Data Sources

Source description: Open Bike Sharing Dataset. The dataset contains information about bike-sharing stations including station name, location, status, available bikes, empty stands, and last update time. The data is used to analyse bike availability, station activity status, and utilization performance.

Domain: Transportation / Urban Mobility Analytics

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3. Problem Statement

- To analysis which stations are actively operating and which are closed
- To analysis bike availability and stand capacity across stations
- To evaluate utilization efficiency of bike stations
- To analysis underperforming locations with low usage
- To understand overall operational performance of the bike-sharing network

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4. Attribute (Column /Features) Details

Attribute Name	Data Type	Description
Number	Integer	Unique station number
Name	Text	Name of the bike station
Address	Text	Station location
Position	Text	Geographic coordinates
Status	Text	Station status (OPEN / CLOSED)
Bike stands	Integer	Total bike stand capacity
Available bike	Integer	Number of bikes currently available
Available bike stands	Integer	Number of empty stands
Last update	Date Time	Last update time of station data

Latitude	Decimal Number	Geographic coordinate that represents the north–south position of the bike station on the Earth's surface. It helps identify the exact location of the station on a map.
Longitude	Decimal Number	Geographic coordinate that represents the east–west position of the bike station. It is used along with latitude to accurately map the station location.

5. Tools & Technologies

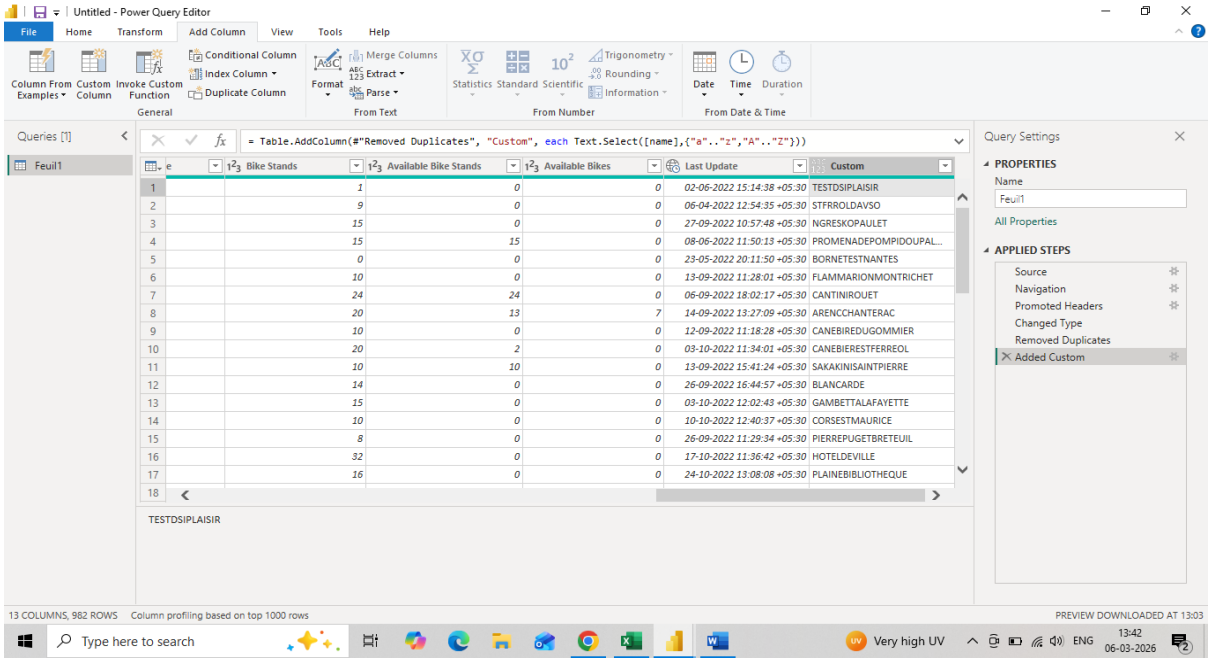
Power BI: Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (Power Query)

Data Cleaning & Transformation:

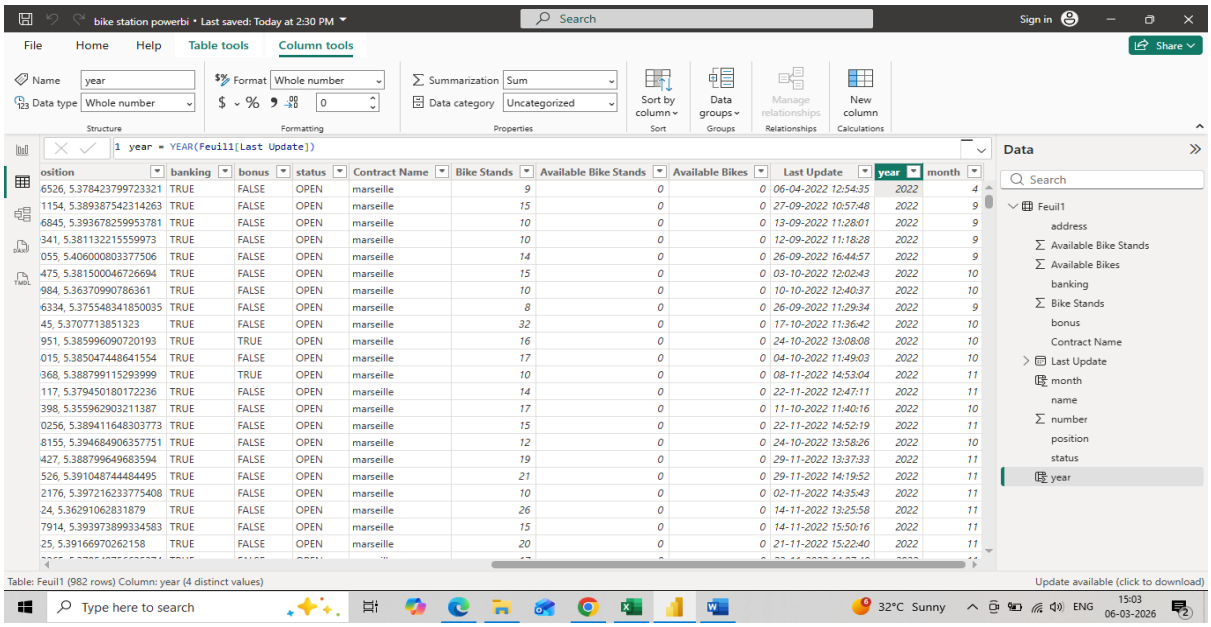
Removed duplicates, handled missing values, standardized formats, and created calculated fields.

1. Custom column created to separate number and name in name column
Name=Text.Select([name],{"a".."z","A".."Z"})



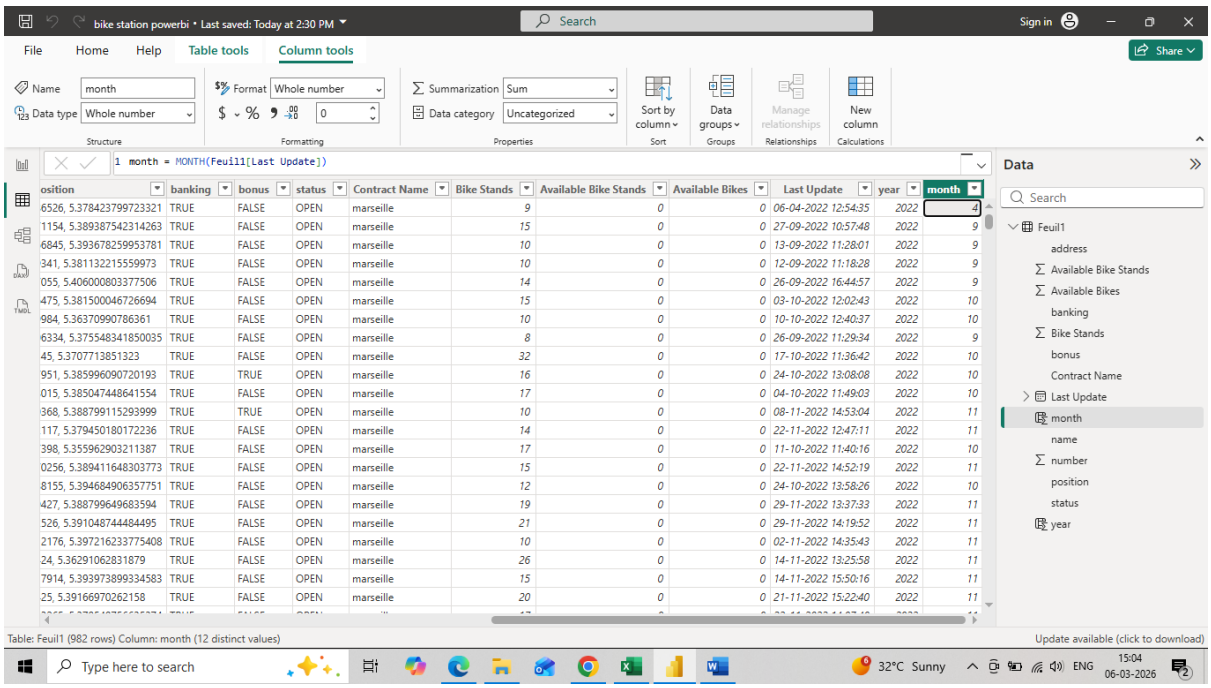
2.Extract year form last update

Year = YEAR(Feuil1[Last Update])

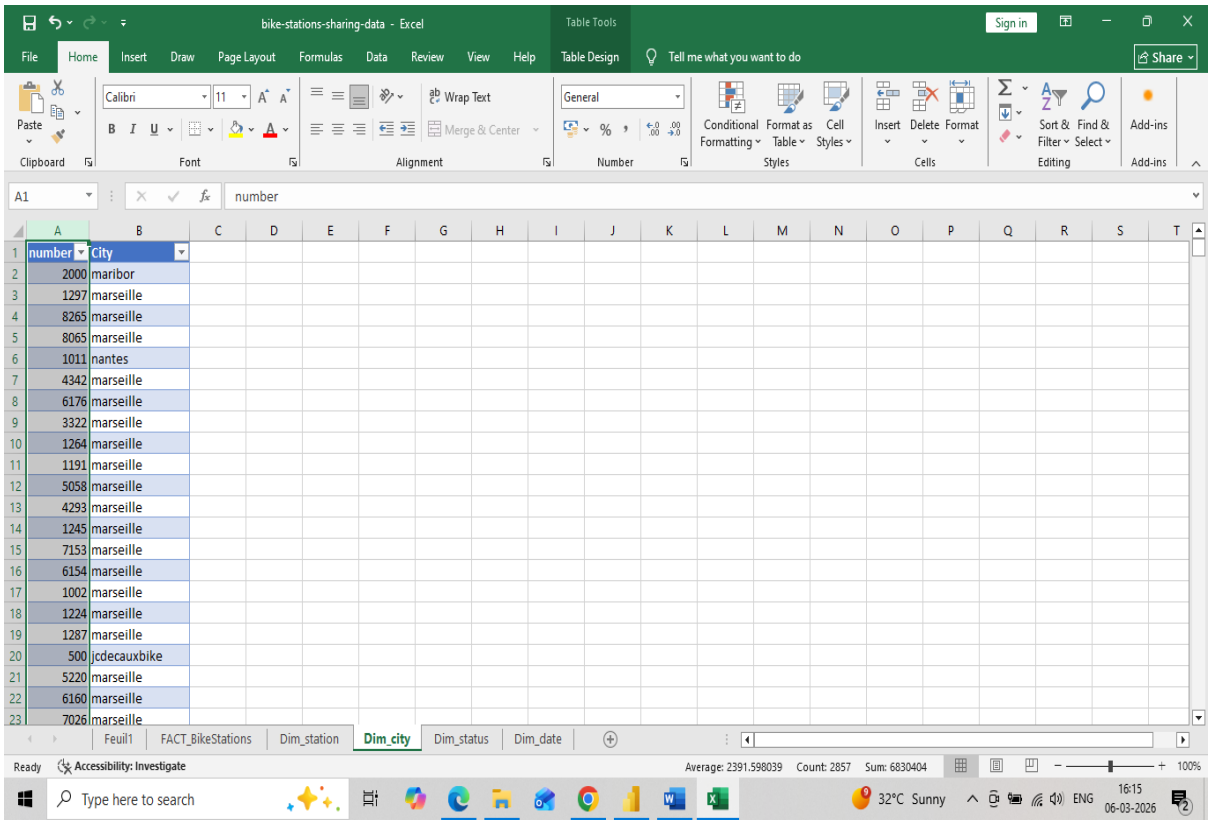


3.Extract month from last update

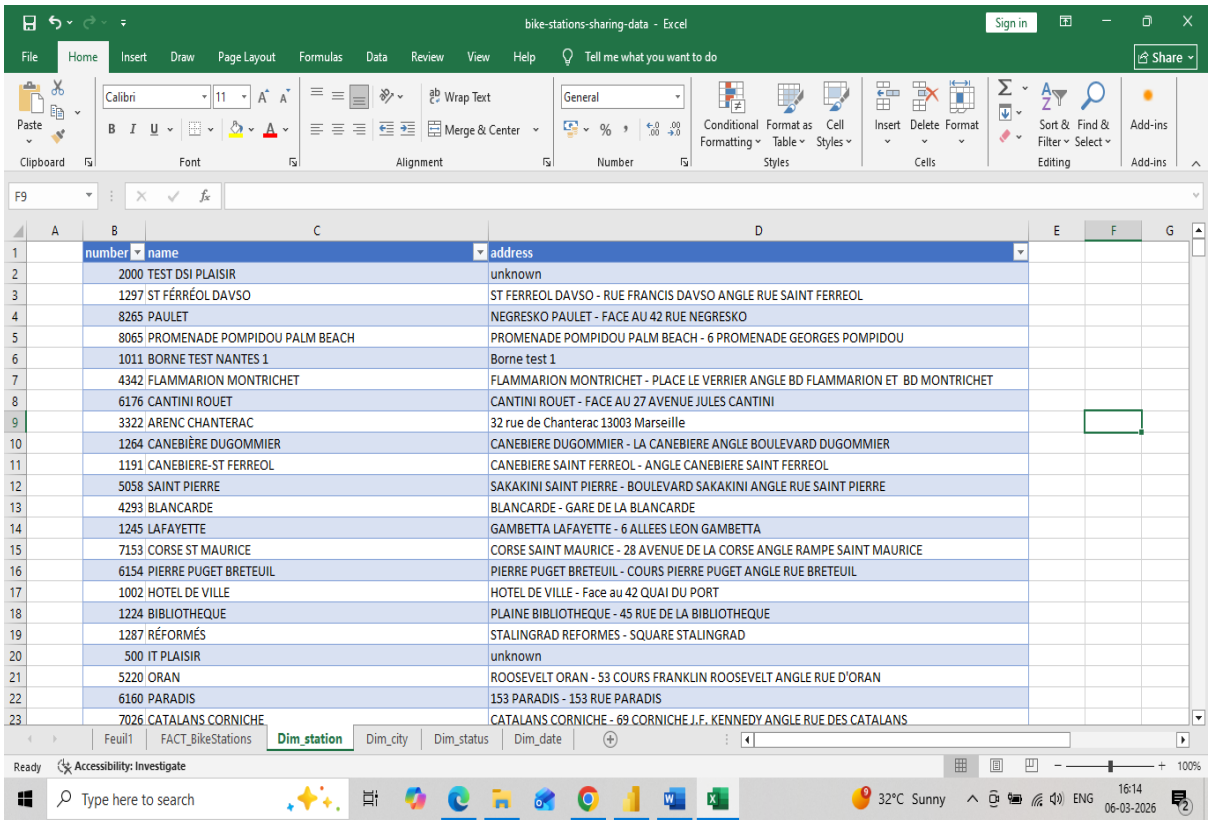
Month = MONTH(Feuil1[Last Update])



Convert the data into Fact and Dimension Table



number	City
2000	maribor
1297	marseille
8265	marseille
8065	marseille
1011	nantes
4342	marseille
6176	marseille
3322	marseille
1264	marseille
1191	marseille
5058	marseille
4293	marseille
1245	marseille
7153	marseille
6154	marseille
1002	marseille
1224	marseille
1287	marseille
500	jcdecauxbike
5220	marseille
6160	marseille
7026	marseille



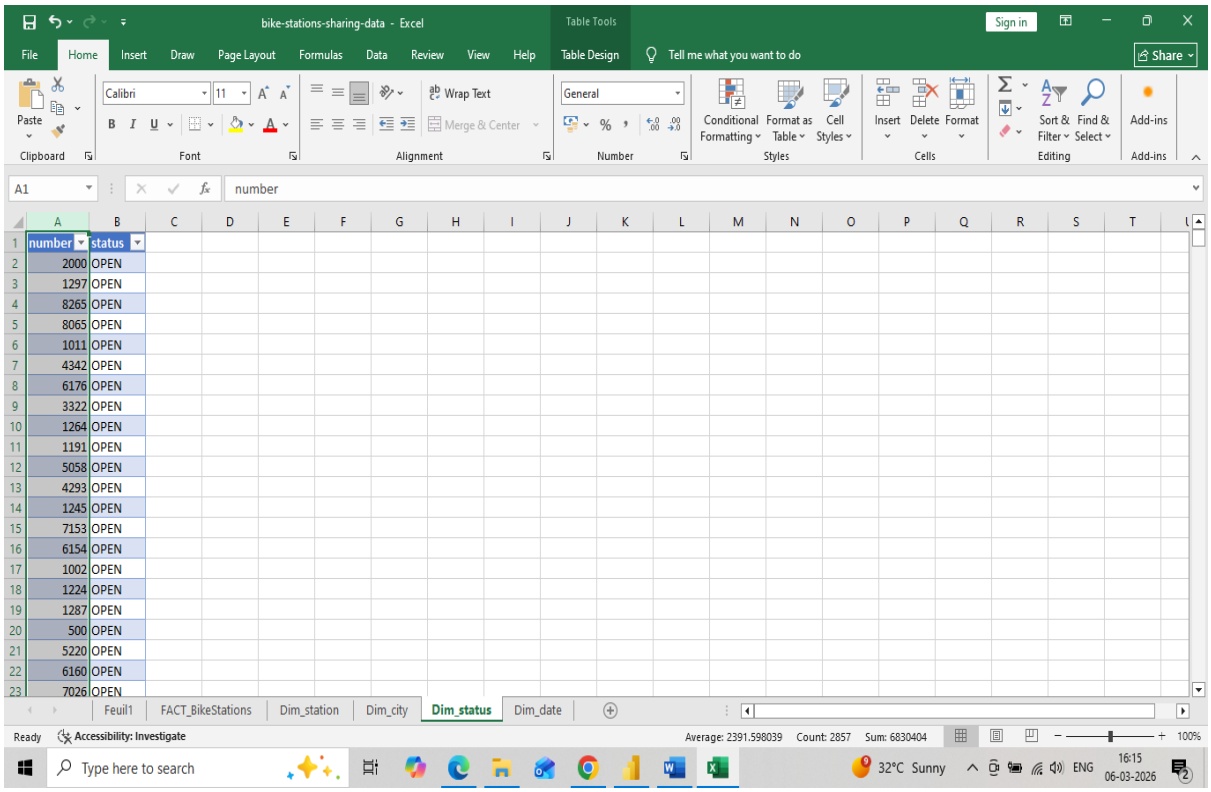
number	name	address
2000	TEST DSI PLAISIR	unknown
1297	ST FÉRRÉOL DAVSO	ST FERREOL DAVSO - RUE FRANCIS DAVSO ANGLE RUE SAINT FERREOL
8265	PAULET	NEGRESKO PAULET - FACE AU 42 RUE NEGRESKO
8065	PROMENADE POMPIDOU PALM BEACH	PROMENADE POMPIDOU PALM BEACH - 6 PROMENADE GEORGES POMPIDOU
1011	BORNE TEST NANTES 1	Borne test 1
4342	FLAMMARION MONTRICHET	FLAMMARION MONTRICHET - PLACE LE VERRIER ANGLE BD FLAMMARION ET BD MONTRICHET
6176	CANTINI ROUET	CANTINI ROUET - FACE AU 27 AVENUE JULES CANTINI
3322	ARENC CHANTERAC	32 rue de Chanterac 13003 Marseille
1264	CANEBIÈRE DUGOMMIER	CANEBIERE DUGOMMIER - LA CANEBIERE ANGLE BOULEVARD DUGOMMIER
1191	CANEBIERE-ST FERREOL	CANEBIERE SAINT FERREOL - ANGLE CANEBIERE SAINT FERREOL
5058	SAINT PIERRE	SAKAKINI SAINT PIERRE - BOULEVARD SAKAKINI ANGLE RUE SAINT PIERRE
4293	BLANCARDE	BLANCARDE - GARE DE LA BLANCARDE
1245	LAFAYETTE	GAMBETTA LAFAYETTE - 6 ALLEES LEON GAMBETTA
7153	CORSE ST MAURICE	CORSE SAINT MAURICE - 28 AVENUE DE LA CORSE ANGLE RAMPE SAINT MAURICE
6154	PIERRE PUGET BRETEUIL	PIERRE PUGET BRETEUIL - COURS PIERRE PUGET ANGLE RUE BRETEUIL
1002	HOTEL DE VILLE	HOTEL DE VILLE - Face au 42 QUAI DU PORT
1224	BIBLIOTHEQUE	PLAINE BIBLIOTHEQUE - 45 RUE DE LA BIBLIOTHEQUE
1287	RÉFORMÉS	STALINGRAD REFORMES - SQUARE STALINGRAD
500	IT PLAISIR	unknown
5220	ORAN	ROOSEVELT ORAN - 53 COURS FRANKLIN ROOSEVELT ANGLE RUE D'ORAN
6160	PARADIS	153 PARADIS - 153 RUE PARADIS
7026	CATALANS CORNICHE	CATALANS CORNICHE - 69 CORNICHE J.F. KENNEDY ANGLE RUE DES CATALANS

bike-stations-sharing-data - Excel

number	Bike Stands	Available Bike Stands	Available Bikes
2000	1	0	0
1297	9	0	0
8265	15	0	0
8065	15	15	0
1011	0	0	0
4342	10	0	0
6176	24	24	0
3322	20	13	7
1264	10	0	0
1191	20	2	0
5058	10	10	0
4293	14	0	0
1245	15	0	0
7153	10	0	0
6154	8	0	0
1002	32	0	0
1224	16	0	0
1287	17	0	0
500	1	0	1
5220	10	0	0
6160	14	0	0
7026	17	0	0

bike-stations-sharing-data - Excel

number	Last Update	Year	Month
2000	2022-06-02T15:14:38+05:30	2022	06
1297	2022-04-06T12:54:35+05:30	2022	04
8265	2022-09-27T10:57:48+05:30	2022	09
8065	2022-06-08T11:50:13+05:30	2022	06
1011	2022-05-23T20:11:50+05:30	2022	05
4342	2022-09-13T11:28:01+05:30	2022	09
6176	2022-09-06T18:02:17+05:30	2022	09
3322	2022-09-14T13:27:09+05:30	2022	09
1264	2022-09-12T11:18:28+05:30	2022	09
1191	2022-10-03T11:34:01+05:30	2022	10
5058	2022-09-13T15:41:24+05:30	2022	09
4293	2022-09-26T16:44:57+05:30	2022	09
1245	2022-10-03T12:02:43+05:30	2022	10
7153	2022-10-10T12:40:37+05:30	2022	10
6154	2022-09-26T11:29:34+05:30	2022	09
1002	2022-10-17T11:36:42+05:30	2022	10
1224	2022-10-24T13:08:08+05:30	2022	10
1287	2022-10-04T11:49:03+05:30	2022	10
500	2022-08-30T21:00:02+05:30	2022	08
5220	2022-11-08T14:53:04+05:30	2022	11
6160	2022-11-22T12:47:11+05:30	2022	11
7026	2022-10-11T11:40:16+05:30	2022	10



number	status
2000	OPEN
1297	OPEN
8265	OPEN
8065	OPEN
1011	OPEN
4342	OPEN
6176	OPEN
3322	OPEN
1264	OPEN
1191	OPEN
5058	OPEN
4293	OPEN
1245	OPEN
7153	OPEN
6154	OPEN
1002	OPEN
1224	OPEN
1287	OPEN
500	OPEN
5220	OPEN
6160	OPEN
7026	OPEN

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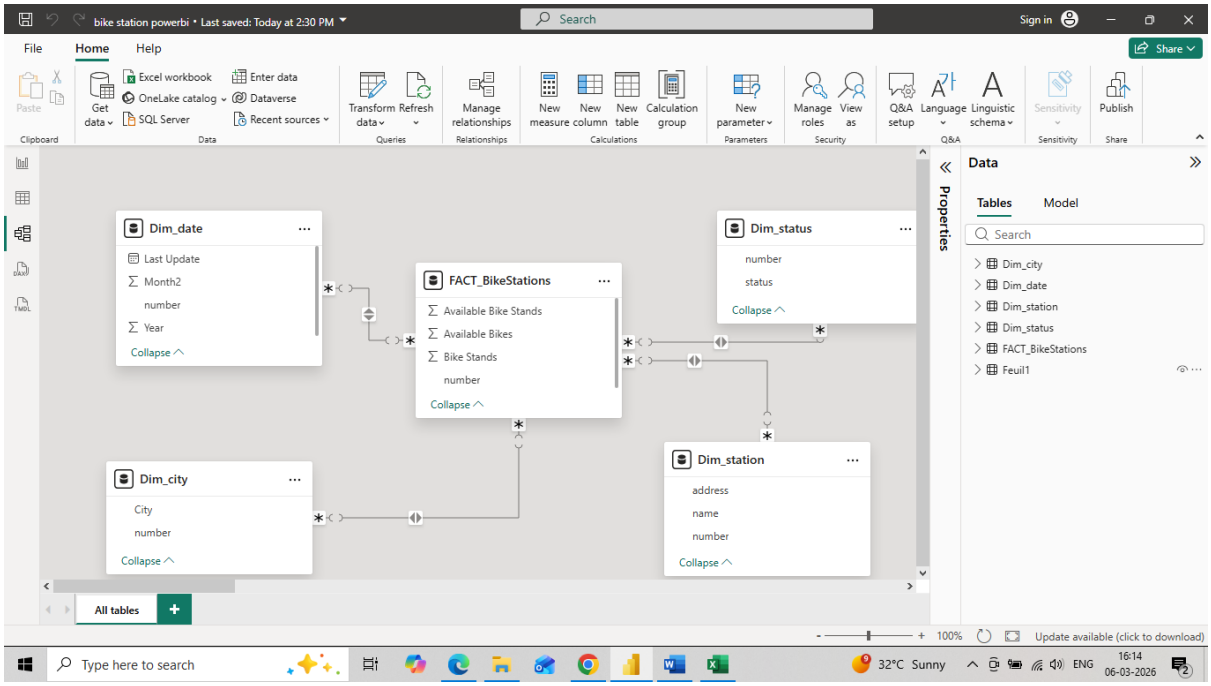
7. Data Modelling and DAX (Power BI)

Data Model: Established relationships between tables, defined cardinality, and created lookup tables where necessary.



- A star schema data model was used to organize the dataset efficiently for analysis in Power BI. The data was structured into Fact and Dimension tables to improve query performance and simplify reporting.
- The Fact Table (FACT_BikeStations) contains the main measurable data related to bike station operations, such as available bikes, bike stands, empty stands, and station status. These numerical values are used for calculations, aggregations, and performance analysis.

- The Dimension Tables contain descriptive information such as station name, address, city, and date details. These tables help categorize and filter the data during analysis.
- By separating the data into fact and dimension tables and creating relationships between them, the model enables efficient data analysis, faster reporting, and better dashboard performance in Power BI.

Calculated Columns & DAX Measures: Implemented DAX formulas for key metrics, such as total sales, growth percentages, and cumulative totals.

Active station = CALCULATE(COUNT(FACT_BikeStations[number]),FACT_BikeStations[status]="OPEN")

Closed_station = CALCULATE(COUNT(FACT_BikeStations[number]),FACT_BikeStations[status]="CLOSED")

Empty_stands = sum(FACT_BikeStations[Available Bike Stands])

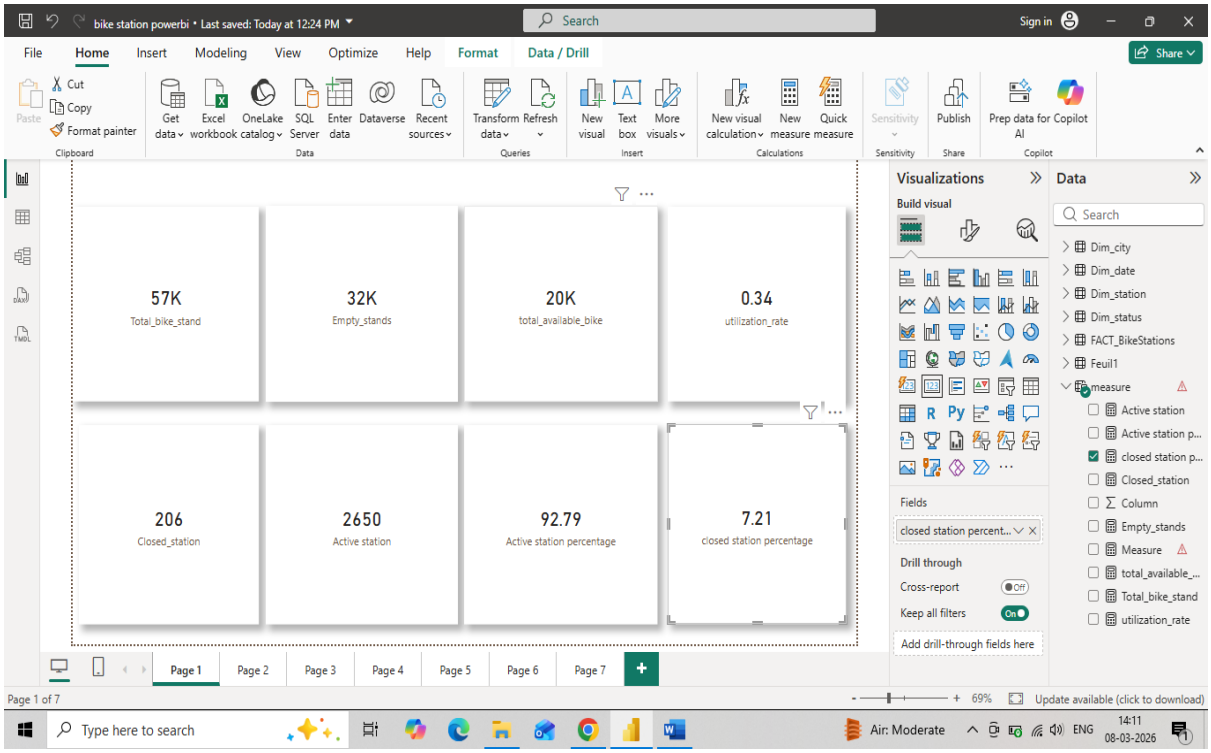
Total_available_bike = sum(FACT_BikeStations[Available Bikes])

Total_bike_stand = sum(FACT_BikeStations[Bike Stands])

Utilization_rate = DIVIDE([total_available_bike],[Total_bike_stand])

Active station percentage = DIVIDE([Active station],[Active station]+[Closed_station])*100

Closed station percentage = DIVIDE([Closed_station],[Active station]+[Closed_station])*100

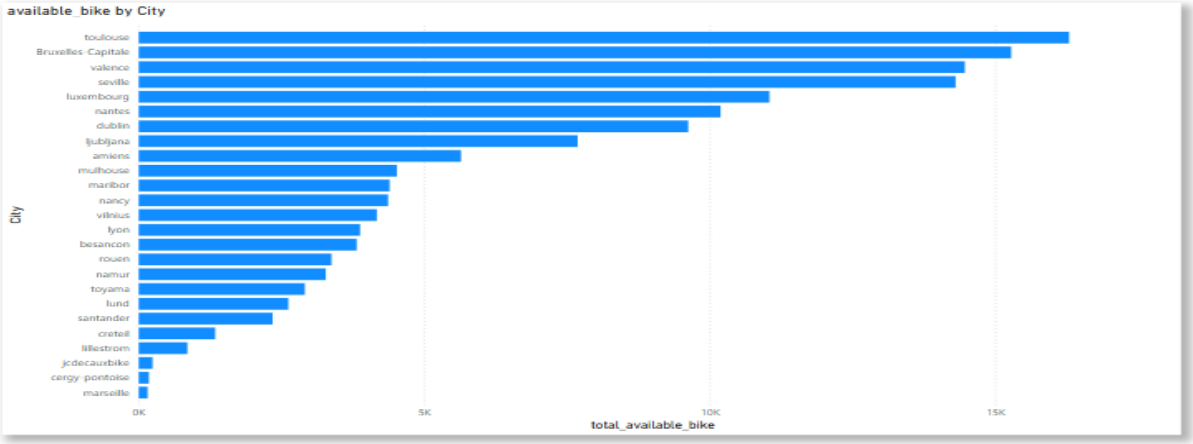


8. Analysis and Visualizations (Power BI)

1.Total Available Bikes by City:

Insights:

This chart shows the bike-sharing network. Toulouse, Bruxelles-Capitale, and Valence have the highest number of available bikes, indicating stronger bike-sharing infrastructure. Cities such as Cergy-Pontoise , Marseille, and Lillestrom show significantly lower bike availability, suggesting limited station capacity or lower system coverage. Understanding bike availability across cities helps operators optimize bike distribution and improve service accessibility in lower-supply areas.



2. Geographical Distribution of Available Bikes (Latitude & Longitude)

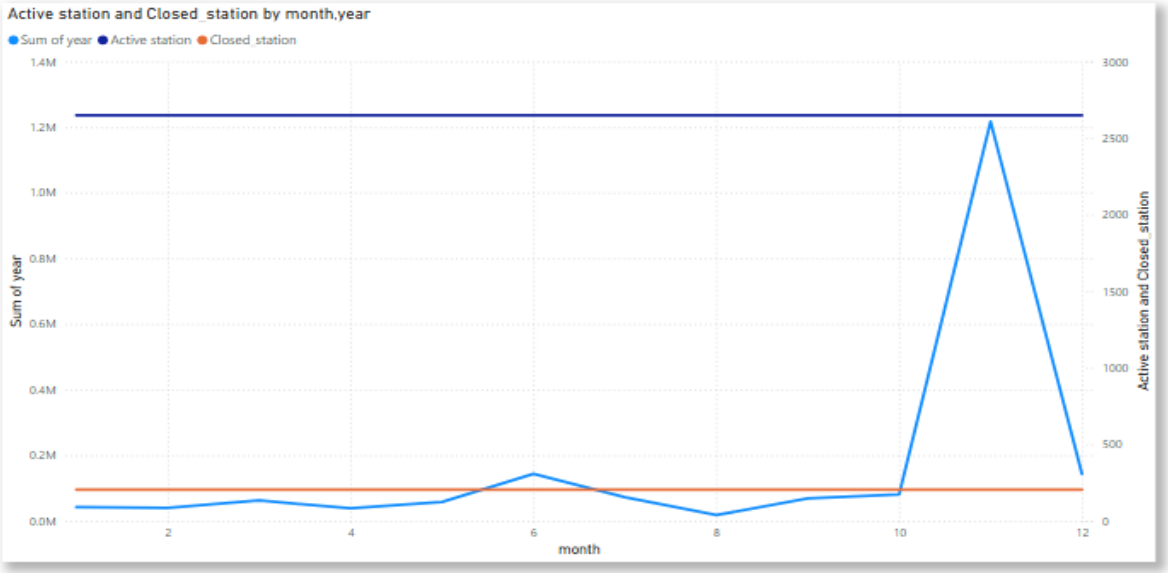
Insights:

The map shows across different cities and countries using latitude and longitude coordinates. Most bike-sharing stations are concentrated in European regions such as France, Belgium, and nearby countries, indicating strong adoption of bike-sharing systems in these areas. A few stations appear in other regions such as Japan, showing the global presence of the bike-sharing network.

3. Monthly Comparison of Active vs. Closed Stations

Insights:

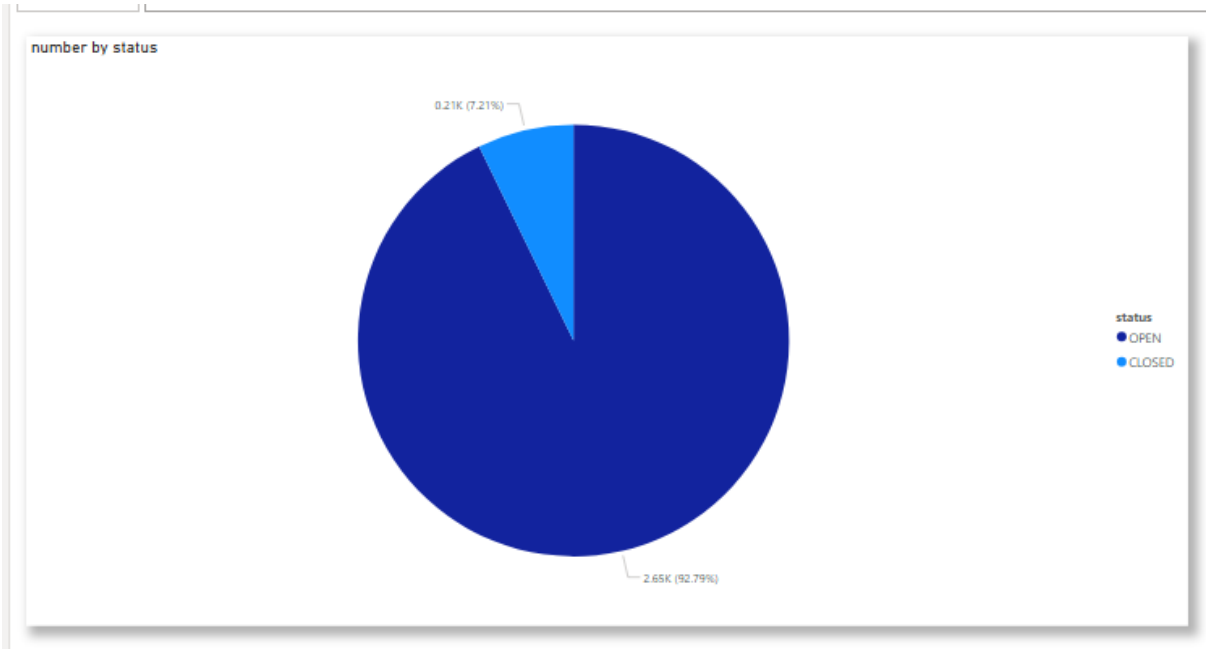
The Line chart shows there is a significant, sharp spike in the "Sum of year" metric during month November, reaching a peak of approximately 1.2M. Both Active stations and Closed stations remain perfectly constant throughout the entire year, showing no monthly fluctuation. Operational Ratio: The number of active stations consistently remains significantly higher than the number of closed stations across all twelve months.



4. Active vs Closed Stations

Insights:

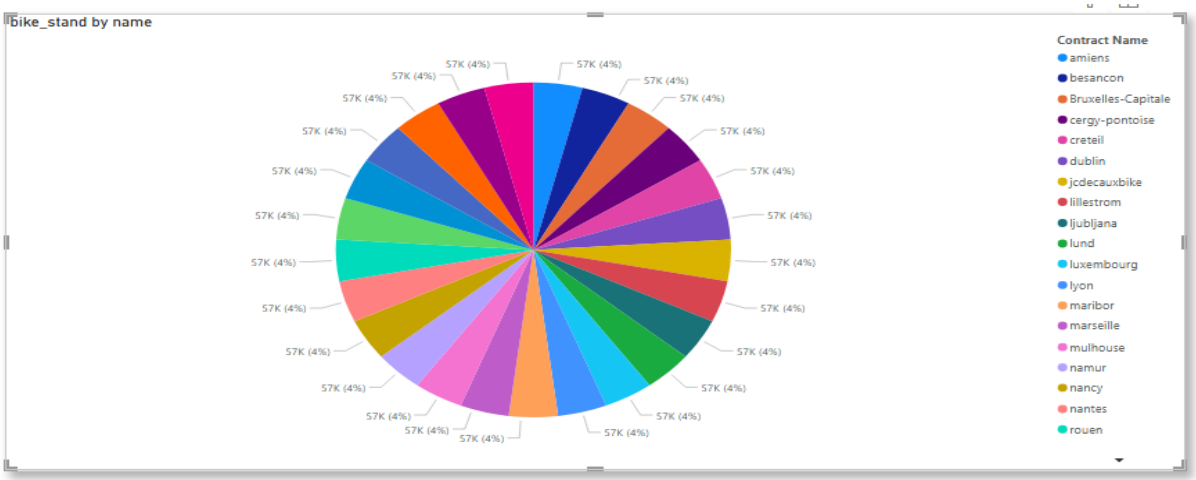
The majority of bike stations are operational, with **92.79% of stations active** and **7.21% closed**. This indicates that the bike-sharing network is largely available for public use. A small portion of stations are closed, which may be due to maintenance or operational issues. Maintaining a high percentage of active stations ensures better accessibility for commuters.



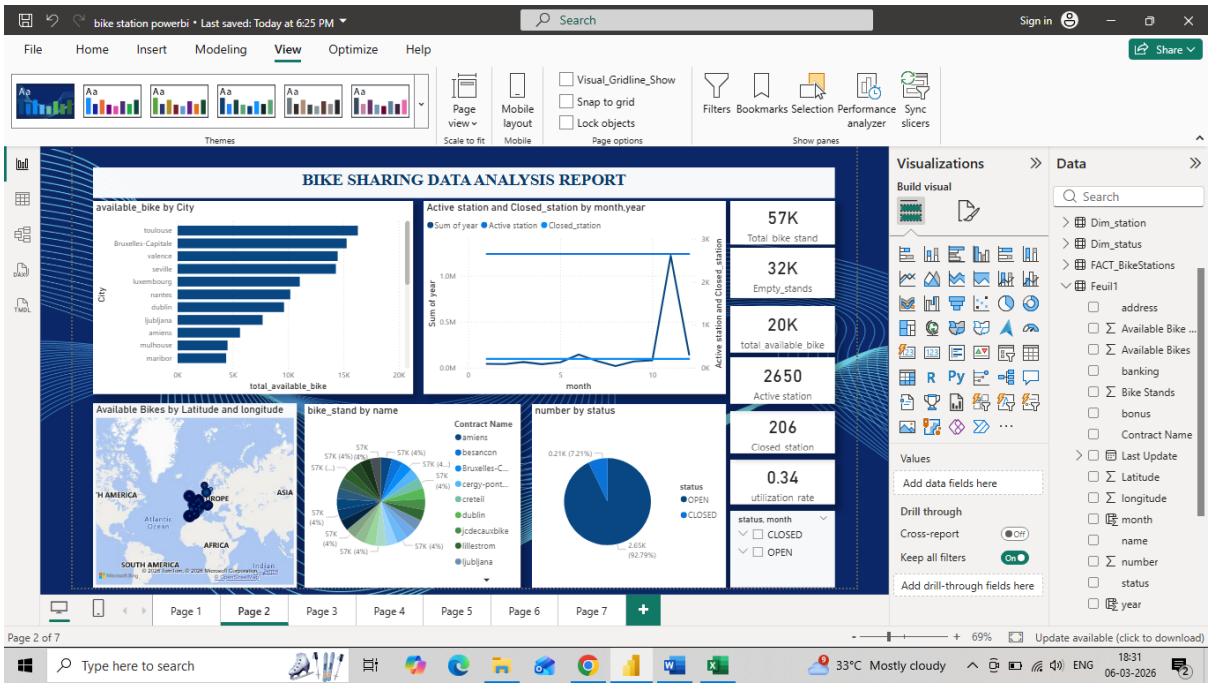
5. Bike Stand Distribution by Contract Name

Insights:

Every contract or city listed in the dataset accounts for an identical 4% share of the total bike stands. Each specific location, such as Dublin, Lyon, and Marseille, contributes exactly 57K stands to the network. The visualization highlights a highly diversified infrastructure spanning across numerous international cities. There is no dominant market leader, suggesting a perfectly balanced and standardized operational model across all regions.



DASHBOARD



9. Summarize Insights

1. Which cities earned the highest and lowest revenue per km?
Although the dataset does not include revenue information, bike utilization rate was used as a proxy to evaluate operational efficiency. Cities such as Lyon and Toulouse show higher utilization rates, indicating better bike usage efficiency, while cities like Rouen and Vilnius show comparatively lower utilization levels.

2. How have revenues changed over the years?
By analyzing the year and month extracted from the Last Update field, the bike-sharing system shows consistent operational activity over time, indicating stable demand and usage patterns across cities.

3. Identify underperforming cities with low cost efficiency.
Cities such as Mulhouse, Cergy-Pontoise , and Rouen show relatively lower bike utilization despite having good bike stand capacity. These cities have higher numbers of available bikes and empty stands, indicating that the existing bike-sharing infrastructure is not being fully utilized.

4. Which cities have the highest bike availability
Cities such as Bruxelles-Capitale and Lyon show the highest number of available bikes, indicating stronger infrastructure and greater bike-sharing accessibility.

5.What percentage of bike stations are active versus closed?

Approximately 92.79% of bike stations are active (OPEN), while 7.21% of stations are closed. This indicates that the majority of the bike-sharing network is operational and available for public use, with only a small portion of stations currently inactive.

6. Which stations show the highest bike utilization?
Stations with fewer available bikes and more empty stands indicate higher demand and frequent usage. Stations located in Lyon and Toulouse, such as central city stations, show higher utilization levels, suggesting that these areas act as important mobility hubs for commuters.

Recommendations:

• Increase bike availability in high-demand cities such as Lyon and Toulouse.
• Improve maintenance and monitoring of closed stations to ensure maximum system availability.
• Redistribute bikes from low-demand areas to high-demand stations to balance supply.
• Expand bike-sharing infrastructure in cities with high utilization rates to support growing demand.

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10. Conclusions

- The integration of Power BI enabled efficient data preparation and visualization for analysing bike-sharing station performance.
- Through interactive dashboards and DAX calculations, meaningful insights were generated regarding station utilization, operational status, and infrastructure efficiency.
- These insights can help transportation authorities improve bike distribution strategies, optimize station locations, and enhance urban mobility services.